

MARSEILLES, IL, USA

WMO: 744600

Lat: 41.367N

Lon: 88.683W

Elev: 225

StdP: 98.65

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-17.6	-24.8	0.4	-19.7	-21.1	0.6	-16.7	12.8	-3.6	12.0	-4.5	3.8	290	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.6	23.6	32.0	23.4	30.4	22.3	25.8	31.1	24.8	30.0	23.8	28.7	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	19.8	29.6	23.1	18.4	27.6	22.1	17.3	27.0	81.1	30.6	76.1	30.4	72.1	28.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.6	-23.3	35.2	3.5	2.1	-25.8	36.7	-27.9	38.0	-29.9	39.2	-32.5	40.7		
(5)			-23.7	27.0	3.5	0.9	-26.1	27.7	-28.2	28.2	-30.1	28.7	-32.6	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.9	-2.9	3.6	9.8	16.0	21.2	23.1	21.9	17.6	10.7	3.8	-2.3		
	DBStd	10.99	6.53	6.38	6.16	5.43	4.94	4.00	3.27	3.89	4.48	5.01	5.41	6.15		
	HDD10.0	1717	432	360	216	67	6	0	0	0	3	54	199	382		
	HDD18.3	3523	689	593	457	262	107	18	2	11	67	239	437	640		
	CDD10.0	1703	0	1	19	61	192	337	405	369	231	77	12	0		
	CDD18.3	467	0	0	1	7	35	106	149	122	45	3	0	0		
	CDH23.3	4577	0	0	12	84	369	1118	1464	1105	386	38	0	0		
	CDH26.7	1654	0	0	2	22	108	424	573	413	107	4	0	0		
(14) Wind	WSAvg	4.2	4.9	4.7	5.1	5.1	4.3	3.6	3.0	2.8	3.2	4.1	4.8	4.5		
(15) Precipitation	PrecAvg	922	55	39	59	82	109	106	108	106	70	74	72	45		
	PrecMax	1231	114	134	115	161	186	215	207	315	159	207	155	76		
	PrecMin	673	11	14	21	39	15	21	7	16	18	9	8	7		
	PrecStd	143	26	28	31	34	51	55	51	69	44	48	40	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	14.3	24.6	29.7	31.4	35.0	35.8	36.6	32.0	27.2	20.8	12.4		
		MCWB	10.0	10.4	16.2	17.6	21.1	23.5	24.3	24.7	23.5	17.5	14.4	8.9		
	2%	DB	8.4	10.5	20.5	25.9	29.7	32.7	33.7	33.2	29.6	24.0	17.4	9.8		
		MCWB	5.7	7.4	14.0	16.8	19.7	22.1	24.0	24.6	21.3	16.0	13.9	7.4		
	5%	DB	5.7	7.4	17.0	22.1	27.5	31.0	31.8	30.8	27.6	21.7	15.0	7.2		
		MCWB	3.6	4.6	12.2	14.7	19.0	21.3	24.1	24.2	19.9	14.6	12.1	5.0		
	10%	DB	3.6	4.9	13.5	19.1	25.1	29.2	30.0	28.7	25.5	19.3	12.4	4.5		
		MCWB	1.9	2.9	9.4	12.8	17.5	20.8	23.2	22.6	18.7	14.0	9.3	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	11.0	17.6	19.2	23.6	24.9	27.2	27.3	24.4	18.6	16.3	10.3		
		MCDB	12.4	13.3	23.3	27.2	29.5	31.2	33.1	32.1	30.4	23.6	17.5	11.6		
	2%	WB	6.2	7.6	15.0	17.6	21.9	23.8	25.9	26.0	22.5	17.5	14.6	7.4		
		MCDB	7.8	10.6	18.8	24.5	27.0	31.0	31.2	31.0	27.8	23.2	17.4	9.6		
	5%	WB	3.7	4.9	12.4	15.7	20.4	22.9	24.9	24.9	21.3	15.8	12.2	5.0		
		MCDB	5.4	7.0	17.3	21.5	25.0	29.3	30.1	29.9	25.7	20.3	15.0	7.1		
	10%	WB	2.1	2.9	9.1	13.8	18.9	21.8	23.8	23.8	20.1	14.3	9.6	3.0		
		MCDB	3.4	4.7	13.5	18.3	23.5	27.5	28.6	27.8	23.9	18.8	12.3	4.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	7.9	9.9	11.7	12.1	11.9	10.9	10.9	11.6	11.2	8.5	7.3		
		MCDBR	10.0	11.3	14.7	15.2	13.9	13.7	12.5	12.3	13.2	14.1	11.0	10.3		
	5% WB	MCWBR	8.2	8.9	9.4	8.9	7.1	6.7	5.8	5.7	6.6	8.1	9.1	8.3		
		MCWBR	9.3	10.7	14.1	13.9	11.3	11.7	11.0	10.6	10.8	12.4	10.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.305	0.340	0.393	0.421	0.430	0.446	0.434	0.403	0.349	0.306	0.288		
	taud		2.438	2.396	2.396	2.255	2.232	2.248	2.190	2.215	2.276	2.433	2.501	2.457		
	Ebn at Noon		871	907	909	877	858	848	829	828	828	844	843	840		
	Edh at Noon		77	94	106	131	138	136	143	135	119	91	73	70		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.58	3.61	4.67	5.35	6.08	6.07	5.36	4.52	3.00	1.95	1.43		
	RadStd		0.17	0.33	0.28	0.38	0.42	0.45	0.34	0.31	0.39	0.34	0.17	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page